



nettest Administrator Guide

Network path troubleshooting with the nettest client and server

Version 0.1 · Covers Windows 11 and Linux · October 2026 (revision 2: service install, scanner banner, serverless probes)

nettest is a two-part tool for diagnosing flaky network paths. The **server** runs at the far end and echoes test traffic. The **client** runs where the problem is seen, measures round-trip latency, packet loss, jitter, connection stability and throughput, and shows the results live in a terminal window or writes them to files for a ticket. Both parts are single executables with no installer and no runtime dependencies.

It speaks the same test protocol over four transports, **WebSocket (ws)**, **WebSocket over TLS (wss)**, **raw TCP** and **UDP**, so you can tell apart "the network drops packets" from "the proxy kills WebSocket connections" from "TLS negotiation is slow" using one tool. For devices that cannot run software, such as IP phones, switches and gateways, the client also probes with **ICMP ping**, **TCP connect**, **SIP OPTIONS** and **HTTP(S)**, with the same statistics, chart and reports (section 8).

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1. Requirements and deployment

Item	Requirement
Operating system	Windows 11 or later (x64). Linux x64 with a current glibc (any mainstream distribution from the last few years).
Dependencies	None. No .NET, no Visual C++ runtime, no OpenSSL, no Python. TLS is built in.
Privileges	Standard user for all tests. <code>ping://</code> uses the unprivileged ICMP API on Windows and the unprivileged "ping socket" on Linux (allowed for everyone on current Fedora, Ubuntu, Debian and other systemd-based distributions; otherwise root). Installing the server as a <i>service</i> needs administrator / root; the installer asks for it itself (section 3).
Disk and memory	About 2.5 MB per executable. Memory stays flat during multi-hour runs.
Terminal	Any modern terminal: Windows Terminal, PowerShell, <code>cmd.exe</code> , or any Linux terminal. The interactive view needs at least 80x24 characters; 130x40 is comfortable.
Network	Server: inbound access on the ports you enable (defaults below). Client: outbound access to those ports, or to the device's ICMP / SIP / HTTP ports for serverless probes. HTTP proxies are not traversed automatically; test the direct path.
Clock sync	Not required. All latency is round-trip and measured on the client's own clock.

Files to deploy

File	Where
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<code>nettest-server.exe / nettest-server</code>	The machine at the far end of the path under test (a server in the datacenter, a cloud VM, the other office).
<code>nettest-client.exe / nettest-client</code>	The machine where the problem is experienced (a user's PC, a branch workstation, a kiosk).

Copy the file anywhere and run it from a terminal. On Linux mark it executable first: `chmod +x nettest-server` . For a permanent server use `nettest-server service install` (section 3), which copies the file to the system location and registers it to start at boot. When run by hand, each side stores its saved settings and, for the server, its TLS certificate, under the per-user configuration directory:

Platform	Configuration directory
Windows	<code>%APPDATA%\nettest\</code> (for example <code>C:\Users\jdoe\AppData\Roaming\nettest\</code>)
Linux	<code>~/.config/nettest/</code>

Firewall. Windows Defender Firewall prompts the first time the server starts; allow it on the network profile you are testing (usually Private or Domain). On Linux, open the ports in `firewalld`, `ufw` or `nftables`. UDP must be allowed inbound on the server like any UDP service. The client needs no inbound rules.

2. Server: settings and operation

Start the server with no arguments for the interactive view, or with `--no-tui` to log to the console (for services, scheduled tasks and SSH sessions). Every setting is available as a command-line switch, as an environment variable, or in the saved configuration file. Precedence is switch, then environment variable, then configuration file, then built-in default.

```
nettest-server
nettest-server --no-tui --token s3cret --log-file C:\logs\nettest-server.log
nettest-server --bind 172.16.89.25 --ws-port 9100 --tcp-port 0 --udp-port 0 --wss-port 0
```

Switch	Environment	Default	Meaning
<code>--bind <ip any></code>	<code>NETTEST_BIND</code>	any	Address to listen on. <code>any</code> binds both IPv4 and IPv6 wildcards. Give a specific IP to restrict to one interface.
<code>--tcp-port N</code>	<code>NETTEST_TCP_PORT</code>	9100	Raw TCP listener. <code>0</code> disables it.
<code>--udp-port N</code>	<code>NETTEST_UDP_PORT</code>	9100	UDP listener. <code>0</code> disables it.
<code>--ws-port N</code>	<code>NETTEST_WS_PORT</code>	9101	WebSocket (<code>ws://</code>) listener. <code>0</code> disables it.
<code>--wss-port N</code>	<code>NETTEST_WSS_PORT</code>	9102	WebSocket over TLS (<code>wss://</code>) listener. <code>0</code> disables it and skips certificate generation.
<code>--token <secret></code>	<code>NETTEST_TOKEN</code>	(none)	Pre-shared token. Clients that do not present the same token are rejected. A foreground server without a token logs a warning at start; a service cannot be installed without one.
<code>--log-file <path></code>		(none)	Append human-readable log lines (connects, disconnects, rejections) to this file.
<code>--jsonl <path></code>		(none)	Append the same events as JSON Lines for log collectors.
<code>--cert-dir <dir></code>		config dir /server/	Where <code>cert.pem</code> and <code>key.pem</code> live. Generated on first start if missing; replace with your own PEM pair if you want a CA-signed certificate.
<code>--idle-timeout N</code>		60	Seconds without traffic before a client session is closed. Must be longer than the client's heartbeat interval in soak tests.
<code>--banner <text></code>	<code>NETTEST_BANNER</code>	nettest by Slaytons Technology Services	Identification text returned to port scanners and browsers on every listener (see "Scanner identification" below). The server version is appended automatically. It cannot be disabled.
<code>--no-tui</code>		off	Plain console logging instead of the interactive view.
<code>--config <path></code>		config dir /server.toml	Use a different configuration file.

<code>--save-config</code>		Write the effective settings to the configuration file, then keep running. Afterwards <code>nettest-server</code> alone starts with those settings.
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Interactive view

The server view shows the active listeners, whether a token is required, the wss certificate fingerprint, a live table of connected clients (protocol, peer address, client id, test mode, connection age, idle time, probe count, bytes in and out) and a scrolling log.

Key	Action
<code>c</code>	Clear the log panel
<code>w</code>	Save the effective settings to the configuration file
<code>?</code>	Help overlay
<code>q</code> or <code>Ctrl-C</code>	Stop the server and exit

TLS (wss) and the fingerprint

When the wss listener is enabled the server creates a self-signed certificate on first start and prints its SHA-256 fingerprint in the log and in the view, for example:

```
wss certificate fingerprint (SHA-256):
6A:52:9B:B2:36:96:47:06:FB:F0:AC:7A:EE:A2:A7:CF:49:9D:89:45:D9:1C:9B:FE:3A:51:BD:52:21:9F:67:ED
```

Give that string to client operators. The client pins it with `--fingerprint` (or the "TLS: pinned SHA-256" field), which proves the client reached the right server even without a CA. The alternative, `--insecure`, accepts any certificate and should only be used on a trusted test network.

Scanner identification

Security scanners (nmap, Nessus, Qualys and the like) must be able to identify every listening port. Anything that connects to a nettest listener without speaking the nettest protocol therefore receives a readable answer:

What connects	What it receives
An HTTP request on the TCP, ws or wss port (a browser, <code>curl</code> , an HTTP scanner)	<code>HTTP/1.1 200 OK</code> with a <code>text/plain</code> body: <code>nettest by Slaytons Technology Services (nettest-server 0.1.0)</code>
Any other bytes, or a connection that sends nothing for 3 seconds (nmap's "null probe")	The same line without HTTP framing, then the connection is closed
A UDP datagram that is not a nettest frame	The same line as one datagram, at most once per source address every 2 seconds and 50 per second overall, so the server cannot be abused to amplify traffic
A TLS client on the wss port that never completes the handshake	The server certificate, which already identifies the service

Real clients are not affected: the first bytes are inspected and then handed to the protocol handler unchanged. Every banner reply is logged as `[scan] <protocol> <peer>: banner sent`, which also makes scan activity visible. To check a server quickly: `curl http://<host>:9100/`.

Security notes

- With a token set, unauthenticated TCP and WebSocket clients receive an error and are disconnected; unauthenticated UDP packets are silently dropped, so the server cannot be used as a traffic reflector. The banner reply is the one exception and is rate-limited as described above.
- A server installed as a service must have a token. Use `--generate-token` to create one and distribute it to client operators; set it on clients through `NETTEST_TOKEN` rather than on the command line where possible.
- The server never executes anything from a client and exposes no files; it only echoes test frames and counts bytes.
- Server exit codes: 0 clean shutdown, 1 interactive view error, 2 could not bind a port, 3 bad arguments or configuration.

3. Server: running as a system service

For a permanent test endpoint the server installs itself as a **systemd unit** on Linux or a **Windows service**, starting at boot with fixed, pre-determined settings. All of this is done by the executable itself; no unit files or `sc.exe` commands need to be written by hand.

```
nettest-server service install --token <secret> # all four
listeners, default ports
nettest-server service install --generate-token --ws-port 9100 --tcp-port 0 --udp-port 0 --wss-
port 0
nettest-server service edit --idle-timeout 120 --wss-port 9102 # change
settings and restart
nettest-server service status
nettest-server service stop | start | restart
nettest-server service uninstall # add --purge to delete settings, certificates and
logs too
```

Subcommand	What it does
<code>install [settings]</code>	Copies the executable to the system location, writes the settings file, registers the service to start at boot and starts it. Accepts every server switch from section 2 plus <code>--generate-token</code> , <code>--no-copy</code> (run from the current location instead of copying) and, on Windows, <code>--account</code> . Re-running <code>install</code> replaces the settings but keeps an existing token unless a new one is given.
<code>edit [settings]</code>	Applies the given switches to the stored settings and restarts the service. Only the switches you give change. The token can be replaced (<code>--token</code> , <code>--generate-token</code>) but not removed.
<code>uninstall [--purge]</code>	Stops and removes the service and the copied executable. <code>--purge</code> also deletes the settings, certificates and log files.
<code>start</code> , <code>stop</code> , <code>restart</code>	Control the installed service.
<code>status</code>	Shows whether the service is installed and running, the stored settings (token masked), the wss fingerprint and where the log is. Needs no elevation.

A token is mandatory. `install` refuses to continue (exit code 3) when no token is given through `--token`, the `NETTEST_TOKEN` environment variable or `--generate-token`. A generated token is printed exactly once, marked "copy it now"; store it in your password manager and give it to client operators. `edit` accepts a new token but never an empty one, so an installed service can never be opened up by mistake.

Administrator rights

Every subcommand except `status` needs root or administrator rights, and the installer obtains them itself. On **Linux** it re-runs through `sudo` (or `pkexec` / `doas` when `sudo` is absent), so the password prompt and all output appear in the same terminal. On **Windows** it triggers the User Account Control prompt; the elevated run happens in the background and its output is shown in your window when it finishes. Declining the prompt ends the command with exit code 3. Running from an already elevated shell skips the prompt.

Where things live

	Linux (systemd)	Windows
--	-----------------	---------

Executable	<code>/usr/local/bin/nettest-server</code>	<code>%ProgramFiles%\nettest\nettest-server.exe</code>
Settings	<code>/etc/nettest/server.toml</code> (root only, mode 0600)	<code>%ProgramData%\nettest\server.toml</code>
Certificates	<code>/var/lib/nettest/</code>	<code>%ProgramData%\nettest\certs\</code>
Log	<code>/var/log/nettest/server.log</code> and <code>journalctl -u nettest-server</code>	<code>%ProgramData%\nettest\server.log</code>
Registration	<code>/etc/systemd/system/nettest-server.service</code>	Service <code>nettest-server</code> ("nettest server" in Services), automatic start, depends on <code>Tcpip</code>
Runs as	A dynamic system user with no shell or home (<code>DynamicUser=yes</code>), read-only system, no privilege escalation. A port below 1024 adds only the bind-privileged-port capability.	<code>LocalSystem</code> by default. <code>--account "NT AUTHORITY\NetworkService"</code> selects a lower-privileged account; grant it write access to <code>%ProgramData%\nettest</code> first.

- The service reads only its own settings file; the per-user `%APPDATA% / ~/.config` files play no part. Settings not given at install time use the defaults from section 2.
- Linux: the settings file reaches the service as a `systemd credential`, which is what lets a root-only file be read by the dynamic user. This requires `systemd 247` (released 2020) or newer; every supported distribution ships a newer one. `systemctl stop` and server shutdown are graceful.
- Windows: the firewall rule is **not** created automatically. Allow `nettest-server.exe` inbound for the listener ports in Windows Defender Firewall (the prompt does not appear for a service). `%ProgramData%` is readable by local users; keep the token out of other locations.
- Upgrading: copy the new executable next to the old one and run `service install` again with no settings; it replaces the binary, keeps the stored settings and token, and restarts.

Typical output of `service status` :

```

service      nettest-server.service: active
config       /etc/nettest/server.toml
bind         any
listeners    tcp 9100  udp 9100  ws 9101  wss 9102
token        set (32 chars)
cert dir     /var/lib/nettest
log file     /var/log/nettest/server.log
idle         60 s
banner       nettest by Slaytons Technology Services (nettest-server 0.1.0)
wss SHA-256
6A:52:9B:B2:36:96:47:06:FB:F0:AC:7A:EE:A2:A7:CF:49:9D:89:45:D9:1C:9B:FE:3A:51:BD:52:21:9F:67:ED
```

4. Client: interactive terminal UI

Run `nettest-client` with no arguments, or with a target URL to pre-fill the form, for example `nettest-client ws://10.0.0.5:9101`. The client moves through three screens:

```
Settings  --(s)-->  Running: stats | live chart | log  --(x)-->  Summary
```

Settings screen

All test settings are edited here. Text fields are edited with `Enter` (type, then `Enter` to commit or `Esc` to cancel). Toggles flip with `Enter` or `Space`; choice fields such as protocol cycle through their options. Invalid values are refused with a message on the status line. Fields that have no effect with the chosen protocol or mode (for example the token with `ping`, or the throughput fields in latency mode) are shown dimmed and their hint says so. The field reference is in [section 5](#).

Running screen

- **Stats panel** (left): sent/received, lost with percentage, late/out-of-order/duplicate counts, in-flight probes, last RTT, min/avg/max, p50/p95/p99, jitter. Soak runs add disconnects, reconnects, uptime, longest outage and session length. Throughput runs show instantaneous and per-direction Mbit/s.
- **Chart** (right): RTT over time as a line, with red markers along the baseline where probes were lost. In throughput mode it plots instantaneous Mbit/s. Keys `1` `2` `3` switch the window between the last 60 seconds, the last 5 minutes and the whole run.
- **Log panel** (bottom): connection events, every failed probe, reconnect attempts and errors. With "Log failures only" off it also lists every successful probe.
- **Header**: target, mode, link state (connected, disconnected, reconnecting #n; for serverless probes "disconnected" means an outage is open) and elapsed time.

Summary screen

Shown when the run ends. It lists the final statistics, connection-phase timings, stability events, and the path of the HTML report if one was written. From here you can run again with the same settings or go back to the settings screen.

Keys

Key	Screen	Action
<code>↑</code> <code>↓</code> <code>Tab</code> <code>Shift-Tab</code>	Settings	Move between fields
<code>Enter</code>	Settings	Edit a text field, or flip a toggle
<code>Space</code>	Settings	Cycle protocol, test mode, throughput direction; flip toggles
<code>s</code> or <code>F5</code>	Settings, Summary	Start the test
<code>w</code>	Settings	Save settings so they are pre-filled next time
<code>x</code> <code>s</code> <code>Esc</code>	Running	Stop the test (waits briefly for in-flight probes so loss is accurate)
<code>r</code>	Running, Summary	Write an HTML report now, from the data collected so far
<code>1</code> <code>2</code> <code>3</code>	Running, Summary	Chart window: last 60 s, last 5 min, everything

c	Running	Clear the log panel
Enter Esc b	Summary	Back to the settings screen
?	Anywhere	Help overlay (any key closes it)
q or Ctrl-C	Anywhere	Quit; a running test is stopped and its report written first

5. Client: settings reference

Each row is one field on the settings screen together with its command-line switch. Switches override saved settings; the saved file is `client.toml` in the configuration directory.

Field	Switch	Default	Meaning
Target host	<code>--host</code> or URL	127.0.0.1	Hostname or IP of the nettest-server, or of any device for the serverless protocols. IPv6 addresses without brackets in the form; with brackets in a URL.
Port	<code>--port</code> or URL	9100	Server port for the chosen protocol; for <code>connect</code> , <code>sip</code> , <code>http</code> , <code>https</code> the device port. Not used by <code>ping</code> . A URL without a port uses the scheme's default (ws 9101, wss 9102, tcp/udp 9100, sip 5060, http 80, https 443).
Protocol	<code>-P</code> , <code>--protocol</code> or URL scheme	ws	Against nettest-server: <code>ws</code> , <code>wss</code> , <code>tcp</code> , <code>udp</code> . Against any device: <code>ping</code> , <code>connect</code> , <code>sip</code> , <code>http</code> , <code>https</code> (section 8).
Test mode	<code>-m</code> , <code>--mode</code>	latency	<code>latency</code> , <code>soak</code> or <code>throughput</code> . See section 7 . Throughput needs a nettest-server; it is refused for the serverless protocols.
Probe interval (ms)	<code>-i</code> , <code>--interval</code>	1000	Time between probes in latency mode. Switch accepts units: <code>200ms</code> , <code>1s</code> . Minimum 1 ms.
Frame size (bytes)	<code>-s</code> , <code>--size</code>	64	Size of each probe on the wire including the 32-byte header. Keep at or below 1400 on UDP to avoid fragmentation. For <code>ping</code> it is the ICMP message size (32 to 65507); the other serverless probes ignore it.
Duration (s)	<code>-d</code> , <code>--duration</code>	0	Stop after this long; 0 means run until stopped. Switch accepts <code>30s</code> , <code>5m</code> , <code>8h</code> .
Probe count	<code>-c</code> , <code>--count</code>	0	Stop after this many probes; 0 means unlimited.
Auth token	<code>--token</code>	(none)	Must equal the server's token when one is configured.
TLS: accept any cert	<code>--insecure</code>	off	wss only. Skip certificate checking. Trusted networks only.
TLS: pinned SHA-256	<code>--fingerprint</code>	(none)	wss only. The server's certificate fingerprint; colons optional. Preferred over "accept any".
Heartbeat (ms)	<code>--heartbeat</code>	5000	Soak mode cadence (also the probe interval for serverless soak tests). Set below any suspected idle timeout to keep the connection alive, or above it to provoke the drop.
Outage after N failures	<code>--outage-after</code>	3	Serverless soak only. This many consecutive failed or unanswered probes open an outage; the next answered probe closes it.
Throughput seconds	<code>--tp-secs</code>	10	Seconds to stream in each direction.
Throughput chunk (bytes)	<code>--tp-chunk</code>	65536	Bytes per throughput frame. Capped to 1368 on UDP automatically.

Throughput direction	<code>--tp-direction</code>	both	<code>upload</code> , <code>download</code> or <code>both</code> (upload first, then download).
Log: CSV file	<code>--csv [path]</code>	off	Write a CSV log. See section 9 .
Log: text file	<code>--text [path]</code>	off	Write a plain-text log.
Log: JSON lines	<code>--jsonl [path]</code>	off	Write a JSON Lines log.
HTML report at end	<code>--report [path] /</code> <code>--no-report</code>	on	Write a single-file HTML report when the run ends.
Log failures only	<code>--all-samples</code> turns it off	on	When on, only failed probes are listed in the log panel, console and text log. CSV and JSONL always contain every probe.
Output directory	<code>-o</code> , <code>--out-dir</code>	<code>.</code>	Where generated files go. Created if missing.
(switch only)	<code>--timeout</code>	5s	Connect timeout per attempt.
(switch only)	<code>-4</code> / <code>-6</code>	auto	Force IPv4 or IPv6 when the host name resolves to both.
(switch only)	<code>--ws-path</code> , <code>--path</code>	<code>/</code>	Request path for WebSocket (useful behind a path-routing proxy) and for <code>http</code> / <code>https</code> probes. A path in the target URL sets it too.
(switch only)	<code>--quiet</code>	off	No console sample or event lines (headless).
(switch only)	<code>--config <path></code>		Use a different settings file.
(switch only)	<code>--save-config</code>		Save the effective settings before running.

Environment variables `NETTEST_HOST` , `NETTEST_PORT` , `NETTEST_PROTOCOL` and `NETTEST_TOKEN` are also honoured, which keeps tokens out of command lines and shell history.

Target shorthand

The first positional argument sets protocol, host and port at once: `wss://edge.example.com:9102` , `udp://10.0.0.5:9100` , `[fd00::5]:9101` , or just `10.0.0.5` . Serverless targets: `ping://10.0.0.77` , `connect://10.0.0.77:5060` (the port is required), `sip://10.0.0.77` , `https://10.0.0.77/admin` .

6. Client: headless and scripted use

Add `--no-tui` to run without the interactive view. The client prints events to the console, writes whatever files are enabled, prints a summary at the end, and returns an exit code that scripts can act on. This is the mode for scheduled tasks, cron, remote shells and monitoring probes.

```
# Five minutes of 100 ms probes over wss, CSV + JSONL + HTML report into C:\runs
nettest-client --no-tui wss://edge.example.com:9102 --fingerprint 6A:52:9B:... ^
-i 100ms -d 5m --csv --jsonl --report -o C:\runs

# Overnight stability check: heartbeat every 5 s for 8 hours, log every disconnect and reconnect
nettest-client --no-tui ws://10.0.0.5:9101 -m soak --heartbeat 5s -d 8h --text

# Throughput, 10 seconds each direction
nettest-client --no-tui tcp://10.0.0.5:9100 -m throughput --tp-secs 10

# Monitoring probe: fail if loss exceeds 1% or p95 exceeds 80 ms, one JSON line on stdout
nettest-client --no-tui udp://10.0.0.5:9100 -i 50ms -c 600 --max-loss 1 --max-p95-ms 80 ^
--json-summary --quiet --no-report

# An IP phone with no server behind it: SIP OPTIONS every 10 s overnight, CSV and report
nettest-client --no-tui sip://10.0.0.77 -m soak --heartbeat 10s -d 8h --csv --report
```

On Linux use `\` instead of `^` for line continuation.

Switch	Purpose
<code>--max-loss <pct></code>	Exit 1 if loss is above this percentage. Default 0, so any loss fails the run.
<code>--max-p95-ms <ms></code>	Exit 1 if the p95 round-trip time is above this value.
<code>--json-summary</code>	Print the final summary as one JSON object on standard output instead of the text table. Everything else goes to standard error.
<code>--quiet</code>	Suppress per-probe and event console output.
<code>--all-samples</code>	Print every probe to the console, not only failures.

Exit codes

Code	Meaning
0	Run completed and loss and p95 were within thresholds.
1	Run completed but loss or p95 exceeded the thresholds.
2	Could not connect, authentication rejected, TLS failed (wrong fingerprint), or a serverless probe could not be set up (name not found, ICMP not permitted).
3	Bad arguments, unreadable configuration, wss without <code>--insecure</code> or <code>--fingerprint</code> , or throughput mode with a serverless protocol.
4	Could not write a log file or report.
5	Interrupted with Ctrl-C before a bounded run (<code>-c</code> or <code>-d</code>) completed.

Ending an unbounded run with Ctrl-C is normal and returns 0 or 1 according to the thresholds.

7. Test modes and what they measure

Mode	What happens	What you get
latency	The client sends numbered probes at a fixed interval; the server echoes each one back.	Round-trip time min/avg/max and p50/p95/p99, jitter, loss percentage, late replies, out-of-order and duplicate counts.
soak	One long-lived connection with periodic heartbeats. If it drops, the client reconnects with increasing back-off and keeps going.	Every disconnect with its reason, reconnect attempts and downtime per outage, uptime percentage, longest outage, longest unbroken session, plus the same RTT statistics from the heartbeats.
throughput	The client streams data to the server for N seconds, then the server streams back for N seconds (or one direction only).	Mbit/s, bytes and frame counts per direction. On UDP, the percentage of datagrams that never arrived.

All modes record the connection phases for every connection attempt: DNS lookup, TCP connect, TLS handshake, WebSocket upgrade and the authentication exchange, each timed separately. Phases that do not apply to the protocol are shown as a dash.

How loss is decided. A probe is counted as lost when no echo has arrived after 2 seconds or ten probe intervals, whichever is longer. If the echo turns up later it is moved from "lost" to "late", so slow links are reported as high latency rather than as loss.

Failed is not lost. When the far end answers negatively before the timeout (TCP connection refused, ICMP host unreachable) the probe is recorded as **failed** with the reason, instead of waiting out the timeout. Failed probes are included in the loss percentage so thresholds behave the same; the summary shows the split.

Round-trip only. RTT is the time from sending a probe to receiving its echo, measured on the client's own clock. The server adds nothing from its clock, so the two machines never need synchronised time. One-way latency is therefore not reported.

8. Testing devices without a server

Many stability complaints concern devices that cannot run nettest-server: desk phones, conference units, switches, access points, printers, gateways. The client tests them directly with a protocol the device already speaks. Pick the scheme in the Protocol field or the target URL; everything else (interval, duration, soak mode, logs, chart, HTML report, exit codes) works as for a nettest-server target.

Scheme	What is sent	What counts as an answer	Notes
ping	ICMP echo request, size from "Frame size" (default 64 bytes).	The matching echo reply.	No port. Unprivileged on Windows and on Linux systems that allow ping sockets (all common distributions); otherwise run as root or use <code>connect</code> . Some devices rate-limit ICMP; keep the interval at 200 ms or slower.
connect	A TCP handshake to the given port, closed immediately with a normal FIN.	Handshake completed. "Connection refused" (host up, port closed) is reported as <i>failed</i> , distinct from a timeout (host or path down).	Port is required (<code>connect://phone:5060</code>). Works against any open TCP port, including the nettest-server ports.
sip	A SIP <code>OPTIONS</code> request over UDP (default port 5060), the standard keep-alive that phones, PBXs and SBCs answer.	Any final SIP response (200 OK, 401, 405, 486...) whose CSeq and Call-ID match the request. The status line is kept as the sample detail.	Tests the phone's SIP stack, not just its IP. Phones that disable <code>OPTIONS</code> show 100% loss; use <code>connect://phone:5060</code> then. SIP over TCP is not probed.
http / https	A <code>HEAD</code> request for the path (default <code>/</code>) with a <code>Host</code> header, timed to the first line of the response.	Any HTTP status line, including 401 and 404 (the web stack is up). The status line is kept as the sample detail.	<code>https</code> accepts any certificate (devices use self-signed ones) unless a fingerprint is pinned. The RTT includes connect and, for https, the TLS handshake.

Stability without a connection

Serverless probes have no connection to lose, so soak mode derives stability from the probes themselves: when **Outage after N failures** (default 3) consecutive probes fail or go unanswered, an outage opens and is logged with the reason; the next answered probe closes it and the downtime is recorded. Disconnect count, uptime percentage, longest outage and longest unbroken session are then reported exactly as for a nettest-server soak, and the header shows "disconnected" while an outage is open. The heartbeat interval is the probe interval in soak mode.

What is the same, what differs

- Same: RTT min/avg/max, percentiles, jitter, loss %, the live chart with loss markers, console/CSV/text/JSONL logs, the HTML report, `--max-loss` / `--max-p95-ms` thresholds and exit codes.
- Connection phases show only DNS (and the peer address); there is no TLS, WebSocket or authentication phase to time.
- A new status **fail** appears in logs and CSV for refused/unreachable answers, with the reason in the detail column. "Late" cannot occur: an answer after the timeout is discarded.

- On Linux an ICMP "destination unreachable" for `ping` is not delivered to unprivileged sockets, so an unreachable host appears as timeouts (as with the `ping` command). Windows reports it as failed.
- Throughput mode is not available: there is nothing on the far end to send or receive bulk data.

Examples

```
# Is the phone there, and how is the path to it? 120 pings, one every half second
nettest-client --no-tui ping://10.0.0.77 -i 500ms -c 120
```

```
# Is its SIP port reachable through the night? Outage after 2 missed probes
nettest-client --no-tui connect://10.0.0.77:5060 -m soak --heartbeat 5s --outage-after 2 -d 10h -
-csv
```

```
# Does the SIP stack answer, and how fast? Status lines in the CSV detail column
nettest-client --no-tui sip://10.0.0.77 -c 60 -i 1s --csv --all-samples
```

```
# The device's web interface, self-signed certificate accepted
nettest-client --no-tui https://10.0.0.77/ -c 20
```

```
# Interactive: pre-filled, then press s
nettest-client sip://10.0.0.77
```

9. Exports and log files

Files are written to the output directory (default: the current directory) and named `nettest-<YYYYMMDD-HHMMSS>.<ext>`. Each switch accepts an optional explicit path instead, for example `--csv run1.csv`.

Output	File	Content
Console	screen	Headless only. One line per failed probe plus every event (connect, disconnect, reconnect, result). <code>--all-samples</code> lists every probe; <code>--quiet</code> silences it.
CSV	<code>.csv</code>	One row per probe <i>and</i> per event. Columns: timestamp, elapsed seconds, kind, sequence, status (ok/lost/late/ooo/dup/fail), RTT ms, running loss %, p50, p95, p99, jitter, detail (failure reason, or the SIP/HTTP status line for serverless probes; event text for events). Opens directly in Excel.
Text log	<code>.log</code>	The same lines as the console, honouring "log failures only".
JSON Lines	<code>.jsonl</code>	One JSON object per probe and event, for Splunk, Elastic, Loki and similar.
HTML report	<code>.html</code>	Single self-contained page: summary tiles, RTT chart with loss markers, throughput table, stability events, connection-phase table and the event log. No JavaScript and no external resources, so it opens offline and is safe to attach to a ticket or e-mail.
Server log	any	<code>--log-file</code> and <code>--jsonl</code> on the server record each client's connect, disconnect reason and per-session totals, plus every <code>[scan]</code> banner reply: the server-side half of the evidence. For a service the log is at the location in section 3.

In the interactive view the HTML report is written automatically when a run ends (if the toggle is on) and can also be written at any moment with `r`. The report path is shown on the status line and the summary screen.

10. Reading the results

Figure	How to read it
last / min / avg / max RTT	Round-trip time in milliseconds. On a LAN expect well under 1 ms; across a WAN tens of ms. Max alone is not alarming; a max far above p99 is a single spike.
p50 / p95 / p99	Percentiles. p95 = 25 ms means 95% of probes returned within 25 ms. These describe what users feel better than the average: a path that is usually fast but stalls once a minute has a fine average and a bad p99.
jitter	Smoothed variation between consecutive RTTs (the estimator used by VoIP). Steady 50 ms is fine for voice; 10 ms that swings by 30 ms is not.
lost / loss %	Probes with no echo. On TCP and WebSocket this is rare because the transport retransmits; loss there shows up as RTT spikes instead. UDP shows true packet loss.
failed	Probes answered negatively (connection refused, host unreachable). Part of the loss count. Many failures with "connection refused" mean the host is up but the service on that port is not.
late	Echoes that arrived after the loss timeout. Many late replies mean a very slow or congested path rather than a lossy one.
ooo / dup	Out-of-order and duplicate echoes. Common on UDP across multi-path links; on TCP they indicate something unusual in the path.

in flight	Probes sent but not yet answered. A climbing value during a run means the path has stalled.
Connect phases	DNS, TCP, TLS, WS upgrade, hello. Whichever is large is where the delay lives: resolver, network, certificate handling, or a proxy.
Soak: disconnects, uptime, longest outage	How often the connection died, how long it stayed dead, and how long it stayed up. Downtime per event is in the HTML report and logs.
Throughput Mbit/s	Payload bits per second on the wire for that direction. Compare upload with download from the same run.

11. Troubleshooting recipes

Symptom	What to run and what it tells you
Application disconnects every few minutes	Soak over the application's protocol with the heartbeat at, say, 30 s. Disconnects at a regular interval point at a NAT, firewall or load-balancer idle timeout. Lower the heartbeat below that interval; if the drops stop, keep-alives are the fix.
Calls or video are choppy	UDP latency test at 20 ms interval for a few minutes. Look at loss %, jitter and p99. Loss and jitter on UDP are what real-time traffic experiences.
"The network is slow"	Latency test over TCP and over ws. Check the connect phases first: high DNS means the resolver, high TCP means the path, high TLS or WS upgrade means a middlebox or certificate issue. Then look at p95 and p99 rather than the average.
WebSocket app fails but the website loads	Run ws and tcp tests to the same server. TCP fine and ws failing with a TCP time but no WS-upgrade time means a proxy or load balancer refuses the upgrade.
Transfers are slow in one direction	Throughput mode with direction "both". A large asymmetry between upload and download usually means traffic shaping or a duplex mismatch.
Loss on UDP but not on TCP	That is genuine packet loss. TCP hides it by retransmitting, which appears as RTT spikes and a high p99 in the TCP test.
Client exits with code 2 immediately	Wrong port or protocol for the server's listener, firewall blocking the port, token mismatch, or a wss fingerprint that does not match. The console line before the summary says which.
A phone keeps losing registration	Run <code>sip://phone -m soak --heartbeat 10s</code> from the same network segment overnight. Outages that coincide with the drops put the fault on the network or the phone; a clean run points at the PBX or the trunk, which you test with <code>connect://pbx:5060</code> or a <code>nettest-server</code> next to the PBX.
Devices on a switch "disappear" briefly	<code>ping://device -m soak --heartbeat 1s --outage-after 3</code> against two or three devices on that switch at once. Simultaneous outages are the switch or its uplink; a single device alone is that device.
<code>ping://</code> reports ICMP not permitted	Linux only: the account's group is outside <code>net.ipv4.ping_group_range</code> . Run as root, widen the range, or use <code>connect://host:port</code> .
A vulnerability scan flags an unknown service on 9100-9102	That is <code>nettest-server</code> ; open <code>http://host:9100/</code> in a browser to see the identification banner. The text is configurable with <code>--banner</code> if the scanner needs a particular string.
Need evidence for a carrier or vendor	Run headless with <code>--csv --report</code> for the agreed period and attach the HTML report; it contains the chart, every event, and the per-phase timings in one file.

12. Quick reference card

Server

nettest-server	# interactive view,
default ports	
nettest-server --no-tui --token <secret> --log-file server.log	# console logging, token required
nettest-server --bind <ip> --ws-port 9100 --tcp-port 0 --udp-port 0 --wss-port 0	# one listener only
nettest-server --save-config	# make current switches the defaults
nettest-server service install --generate-token [--ws-port N ...]	# permanent service (token required)
nettest-server service edit --idle-timeout 120	# change a setting, restart
nettest-server service status start stop restart uninstall	# manage it

Default ports: tcp 9100, udp 9100, ws 9101, wss 9102. Port 0 disables a listener. Every port answers scanners with the banner.

Client

nettest-client ws://<host>:9101	# interactive, pre-filled target
nettest-client --no-tui tcp://<host>:9100 -i 200ms -c 300 --csv	# 300 probes, CSV + HTML report
nettest-client --no-tui ws://<host>:9101 -m soak --heartbeat 5s -d 8h	# overnight stability
nettest-client --no-tui tcp://<host>:9100 -m throughput --tp-secs 10	# bandwidth both directions
nettest-client --no-tui wss://<host>:9102 --fingerprint <sha256>	# TLS with pinned certificate
nettest-client --no-tui ping://<device> -c 100	# no server: ICMP
nettest-client --no-tui sip://<phone> -m soak --heartbeat 10s -d 8h	# no server: SIP OPTIONS stability
nettest-client --no-tui connect://<device>:<port> -m soak	# no server: TCP port reachability

Interactive keys

s	start / run again
x	stop
r	write HTML report
w	save settings
1 2 3	chart window 60 s / 5 min / all
c	clear log

Exit codes

0	ok
1	thresholds exceeded
2	connect / auth / TLS
3	usage / configuration
4	file write failed
5	interrupted

?	help
q	quit

Files

Settings: %APPDATA%\nettest\ or ~/.config/nettest/
Service: %ProgramData%\nettest\ or /etc/nettest/ ,
/var/lib/nettest/ , /var/log/nettest/
Outputs: nettest-<timestamp>.csv / .log / .jsonl /
.html